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Mr. Shubham Raina, SWE III at Google India

Submitted by

Nithin S

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DECLARATION BY THE STUDENT

I, Nithin S, hereby declare that the Practical Training work entitled “Search Ads Quality Model” was carried out by me from **12th May 2025** to **1st August, 2025** of duration **12 weeks** during summer term of the academic year 2024-2025. I declare that this is my original work and has been completed successfully at **Google India** and as per the specifications of NITK Surathkal.

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Google IT Services India Private Limited
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Registered Address

Bagmane Constellation Business Park, 11th-12th Floor, Carina-West Tower, Bangalore, 560048, India

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Shraddhanjali Rao
Authorised Signatory
Visa Letters: visaletters@google.com
Verification Letters: verifications@google.com

ABSTRACT

The problem of displaying irrelevant or low-quality advertisements against user queries remains a critical challenge in digital advertising platforms. This project aimed to design and implement a scalable, learning-based retrieval system that eliminates “bad ads” by leveraging dual encoder architectures and fine-tuned large language models (LLMs). The core idea is to represent both user queries and ad candidates in a shared embedding space, enabling fast similarity computation via dot products. To optimize real-time serving efficiency, ad embeddings were precomputed using features such as URL, headlines, creative text, and location, and stored in a distributed key–value store. For incoming user queries, embeddings were generated in real time and compared against candidate embeddings to retrieve the most relevant ads.

The system was trained on approximately 300 million data samples, ensuring robustness across diverse domains and geographies. A percentile-based scoring mechanism was applied to dynamically adjust thresholds for filtering poor matches. Experiments demonstrated significant improvements in filtering precision, scalability, and latency performance. This report details the methodology, architecture, results, and future directions of the project.

Keywords— Ad Filtering, Dual Encoder, Large Language Models, Semantic Embeddings, Query-Ad Relevance, Percentile Thresholding, Key–Value Store, FAISS, Real-time Inference, Scalable Retrieval Systems.

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1. INTRODUCTION

The relevance of advertisements to user queries is the cornerstone of any online advertising platform. When irrelevant or low-quality ads are displayed, users experience dissatisfaction, which leads to reduced engagement and ultimately affects both advertiser confidence and platform credibility. Traditional approaches to ad matching relied heavily on keyword overlap or manually curated rules, which failed to capture the semantic richness of natural language queries.

With the rapid progress in Natural Language Processing (NLP) and the rise of Large Language Models (LLMs), it has become possible to generate dense, semantically meaningful representations (embeddings) of both queries and ads. Embeddings allow a query such as “cheap hotels in Bangalore” to be matched with ads for “affordable stays in Bengaluru”, despite having minimal lexical overlap. This paradigm shift ensures that ads are retrieved based on their semantic intent rather than surface-level keyword similarity.

This project was designed to leverage these advancements and create a dual encoder-based retrieval system for filtering out bad or irrelevant ads. The system precomputes embeddings for ad candidates and stores them in a distributed infrastructure, while query embeddings are generated dynamically at runtime. By applying a percentile-based filtering strategy, the system ensures only ads with sufficiently high semantic similarity to the query are surfaced. The outcome is a robust, scalable, and real-time ad retrieval system capable of handling massive volumes of data.

1.1 Scope of Work

The scope of this project extended beyond the design of a single model and encompassed the full lifecycle of a real-world ad retrieval system. The key elements included:

- **Model Development:** Design and fine-tuning of a dual encoder architecture using a Large Language Model (LLM) as the backbone to capture semantic similarity between queries

and ads.

- **Embedding Infrastructure:** Precomputing embeddings for hundreds of millions of ads using metadata such as URLs, ad headlines, creative text, advertiser information, and location signals.
- **Efficient Storage and Retrieval:** Building a distributed key–value store capable of supporting low-latency lookups at scale, ensuring embeddings can be retrieved and compared within milliseconds.
- **Similarity Computation:** Implementing dot product–based similarity scoring to compare query embeddings against ad embeddings, optimized through libraries like FAISS for vector similarity search.
- **Thresholding Strategy:** Moving beyond static thresholds by employing percentile-based thresholds that adapt to domain- or market-specific data distributions.
- **Evaluation and Metrics:** Establishing rigorous evaluation metrics, including filtering precision, latency, and scalability, to benchmark the system against legacy methods.

The scope also incorporated **deployment considerations**, ensuring that the designed solution could scale to production environments with billions of ad candidates, while maintaining high availability and robustness under heavy query loads.

1.2 Product Scenarios

The designed system has broad applicability across multiple advertising scenarios. A few representative examples include:

1. E-commerce Queries

- *Example:* A user searches for “budget wireless earbuds.”
- *Expected Behavior:* The system retrieves ads for affordable earphones from trusted vendors while suppressing unrelated or premium-only product ads.

2. Local Business Searches

- *Example:* A query like “best pizza near Koramangala” should return ads only from restaurants located in or around the specified locality.
- *Impact:* Increases ad relevance by incorporating **geographic embeddings** into the retrieval system.

3. Policy-sensitive Queries

- *Example:* A query involving “unauthorized medicines” or restricted categories.
- *Expected Behavior:* No ads should be returned, demonstrating the system’s ability to enforce platform safety policies.

4. Multi-intent Queries

- *Example:* “cheap hotels with free Wi-Fi near the airport.”
- *System Response:* Ads are retrieved only if they match **all aspects** of the intent (affordability, amenities, and location), showing the capability of embeddings to capture nuanced intent.

5. Cold-start Scenarios

- For newly onboarded advertisers, embeddings are generated quickly from available metadata, ensuring their ads are still discoverable without manual intervention.

By covering these scenarios, the system not only improves end-user satisfaction but also enhances advertiser ROI, strengthens compliance with platform policies, and ultimately delivers a more trustworthy ad ecosystem.

2. REQUIREMENT ANALYSIS

Requirement analysis is a crucial step in any software engineering project. It involves identifying the capabilities the system must provide (functional requirements) and the constraints/qualities under which the system must operate (non-functional requirements). Since the goal of this project was to build an intelligent, scalable, and real-time ad filtering system, the requirements needed to balance accuracy of ad retrieval with efficiency at scale.

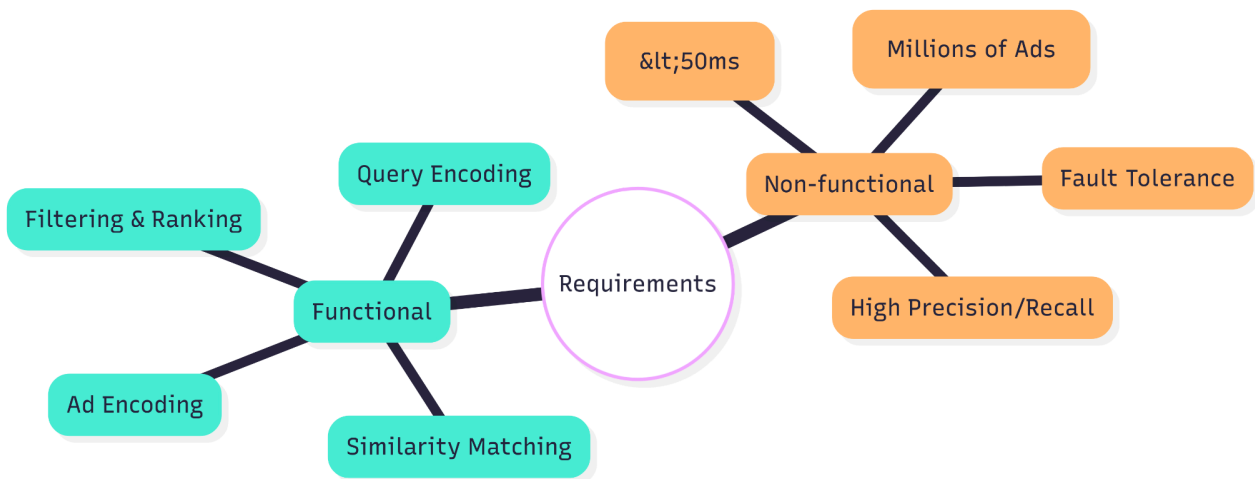


Fig 2.1 : Requirement Analysis

2.1 Functional Requirements

Functional requirements define the specific **operations and services** the system should provide. For the ad filtering system, these were identified as follows:

1. Precomputation of Ad Embeddings

- The system must be able to generate embeddings for all ad candidates from their associated metadata (e.g., URL, ad headlines, creative text, advertiser information, and geographic location).
- These embeddings should be computed offline and stored in a format suitable for fast lookup.

2. Real-time Query Embedding Generation

- For every incoming user query, the system must encode the query into a dense

vector representation using the fine-tuned LLM encoder.

- This process should be optimized to handle high query throughput in real-time environments.

3. Efficient Retrieval of Candidate Embeddings

- The system must support fast retrieval of precomputed ad embeddings from a distributed key–value store.
- Retrieval should support sharding and replication for scalability and reliability.

4. Similarity Scoring

- The system must compute similarity between the query embedding and ad embeddings using dot products.
- It should be capable of handling millions of similarity computations per second, potentially optimized through vector search libraries such as FAISS.

5. Dynamic Thresholding

- Instead of using a fixed similarity cutoff, the system must apply percentile-based thresholds that adjust according to the distribution of similarity scores in the dataset.
- This ensures better adaptability across domains (e.g., e-commerce queries vs. service-related queries).

6. Filtering and Ranking of Ads

- Ads below the threshold must be filtered out, ensuring irrelevant or “bad” ads do not appear to users.
- Ads above the threshold should be ranked based on similarity score or additional signals (e.g., quality score, advertiser bids).

7. Logging and Monitoring

- The system must maintain detailed logs of queries, retrieved ads, and filtering

decisions.

- This data is critical for debugging, performance monitoring, and re-training future versions of the model.

8. Policy Enforcement

- The system must be able to recognize and block policy-sensitive or restricted content categories (e.g., adult content, unauthorized medicines, or harmful products).
- This ensures compliance with regulatory requirements and platform policies.

2.2 Non-Functional Requirements

Non-functional requirements define the **quality attributes** and **operational constraints** of the system. For this project, these were particularly important given the large-scale, real-time nature of ad retrieval.

1. Performance

- Query processing must occur within **50 milliseconds** end-to-end, even under peak traffic loads.
- Vector similarity search operations must be highly optimized for speed without sacrificing accuracy.

2. Scalability

- The system should be able to scale horizontally across multiple servers or clusters.
- It must support billions of ad embeddings and millions of concurrent queries without degradation in performance.

3. Reliability and Fault Tolerance

- The distributed key–value store must ensure redundancy through replication, so no single point of failure can disrupt the system.
- Automatic failover mechanisms should guarantee continued service availability.

4. Consistency

- Embeddings stored in the key–value store should remain consistent across replicas.
- The system must ensure that updates (e.g., new ads or modified metadata) are propagated reliably.

5. Maintainability

- The system should be modular and easy to maintain, with clear separation between components (e.g., model inference, embedding storage, and retrieval).
- Updating or fine-tuning the LLM encoder should require minimal changes to downstream infrastructure.

6. Security and Privacy

- User queries and ad data must be handled securely, with encryption in transit and at rest.
- Access to the system should be restricted through authentication and authorization mechanisms.

7. Adaptability

- The system should allow seamless integration of new models or algorithms (e.g., cross-encoders, multimodal encoders).
- It should also support online re-training pipelines for continuous improvement.

8. Monitoring and Observability

- Real-time dashboards should provide visibility into query volumes, latency metrics, error rates, and filtering accuracy.
- Automated alerts must be triggered in case of anomalies such as sudden latency spikes or retrieval failures.

9. Cost-efficiency

- Infrastructure costs should remain sustainable, especially given the scale of data.
- Use of GPU clusters, distributed storage, and high-throughput search engines must be balanced with optimization strategies to reduce expenses.

10. Compliance

- The system should adhere to regional laws and platform policies regarding advertising (e.g., GDPR for data privacy, local ad regulations).

Requirement Traceability Matrix (RTM)

Requirement ID	Functional Requirement	System Component(s) / Module(s)
FR-1	Precomputation of Ad Embeddings from metadata	Ad Encoder (LLM), Embedding Generation Pipeline, Offline Batch Processing
FR-2	Real-time Query Embedding Generation	Query Encoder (LLM), Inference Service API
FR-3	Efficient Retrieval of Candidate Embeddings from storage	Distributed Key–Value Store, Sharding & Replication Layer
FR-4	Similarity Scoring via dot product computations	Similarity Engine, FAISS Vector Search Library
FR-5	Dynamic Thresholding using percentile-based score distributions	Thresholding Module, Data Distribution Analyzer
FR-6	Filtering and Ranking of Ads	Filtering Engine, Ranking Layer (with quality signals & similarity scores)

FR-7	Logging and Monitoring of queries and retrieval outcomes	Logging Service, Monitoring Dashboards (Prometheus, Grafana)
FR-8	Policy Enforcement to block restricted or unsafe ads	Policy Filter Module, Compliance Rules Engine

Table 2.1 : Requirement Traceability Matrix

Explanation of RTM

- Each **Functional Requirement** has been assigned a unique **Requirement ID** (FR-1, FR-2, ...).
- The **System Component(s)** column shows exactly which modules of the architecture are responsible for satisfying the requirement.
- This helps in validating that **every requirement is implemented** and provides a clear reference for both developers and stakeholders.

2.3 Use Case Scenarios

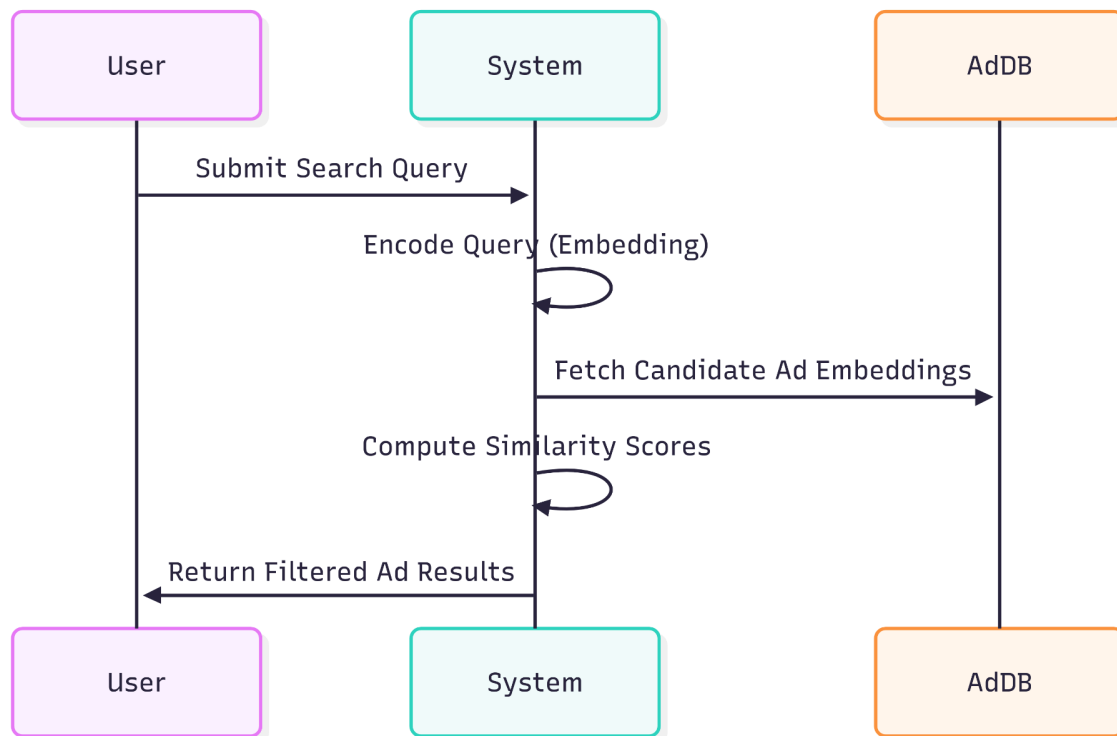


Figure 2.2 : Use Case Scenarios

The designed ad filtering system is expected to serve a wide range of real-world use cases in digital advertising. Each use case highlights the system’s ability to adapt to different query intents, advertiser needs, and platform policies.

Use Case 1: E-commerce Query Matching

- **Scenario:** A user searches for “affordable gaming laptops under ₹60,000.”
- **System Behavior:**
 - The query is encoded into a dense embedding.
 - Candidate embeddings from ads are retrieved, and similarity scores are computed.
 - Ads that mention gaming laptops within the specified budget are surfaced.
 - Irrelevant ads, such as those for unrelated electronics (e.g., mobile phones), are filtered out.

- **Impact:** Users are presented with highly relevant product ads, increasing click-through rate (CTR) and conversion probability.

Use Case 2: Local Business Search

- **Scenario:** A user types “dentist near MG Road, Bangalore.”
- **System Behavior:**
 - The query embedding incorporates **location context**.
 - Ads for dental clinics within the geographic radius are retrieved.
 - Clinics outside the target area are filtered out automatically.
- **Impact:** Enhances **local business discoverability** and improves customer satisfaction by ensuring location-aware ad relevance.

Use Case 3: Policy-sensitive Queries

- **Scenario:** A user searches for “buy prescription drugs without a doctor.”
- **System Behavior:**
 - The query is matched against policy filters.
 - Even if ads exist in the embedding space, the **Policy Enforcement Module** blocks results.
- **Impact:** Maintains compliance with platform safety guidelines, protecting both the company’s reputation and user trust.

Use Case 4: Multi-intent Queries

- **Scenario:** A user enters “cheap hotels with free breakfast near airport.”
- **System Behavior:**
 - The embedding captures multiple semantic dimensions (affordability, amenities, location).
 - Only ads that satisfy all dimensions of intent are ranked highly.
 - Ads failing to meet any one condition (e.g., expensive hotels, missing breakfast) are suppressed.

- **Impact:** Demonstrates the **semantic power of LLM embeddings**, going beyond keyword matching.

Use Case 5: Cold-start for New Advertisers

- **Scenario:** A new advertiser uploads metadata for a “new smartphone launch.”
- **System Behavior:**
 - Embeddings are precomputed immediately from metadata (headlines, URLs, brand descriptions).
 - Ads are ready to be served against relevant queries without manual curation.
- **Impact:** Reduces **onboarding friction** for advertisers and ensures rapid integration into the ecosystem.

Use Case 6: Multilingual Queries

- **Scenario:** A user searches in Hindi (“सस्ते जूते ऑनलाइन”) or Kannada (“ಅಗ್ಗದ ಶೂಸ್”).
- **System Behavior:**
 - The fine-tuned LLM supports multilingual embeddings.
 - Ads in English or local languages are still matched correctly, ensuring inclusivity.
- **Impact:** Expands reach across diverse linguistic user bases in India and globally.

2.4 Other Software Engineering Methodologies

To ensure the system was not only functional but also maintainable and scalable, several software engineering methodologies were adopted during the project. These methodologies supported structured development, testing, and deployment practices.

Agile Development Methodology

- The project followed an **Agile model** with two-week sprints.
- Regular sprint reviews ensured iterative progress on embedding models, infrastructure design, and system evaluation.
- This methodology provided flexibility to adapt based on intermediate findings, such as tuning the percentile thresholds or experimenting with alternate similarity measures.

Continuous Integration and Continuous Deployment (CI/CD)

- A robust **CI/CD pipeline** was employed to automate the building, testing, and deployment of the system.
- Model training updates were pushed through controlled pipelines to avoid regressions.
- Unit tests, integration tests, and automated quality checks were part of each release cycle.

Test-driven Development (TDD)

- Critical modules, such as the **query encoder API** and **similarity computation engine**, were developed using TDD principles.
- Writing tests before implementation helped maintain reliability and ensured the system behaved as expected under edge cases (e.g., empty queries, malformed ad metadata).

Data-driven Methodology

- The availability of **300 million training samples** necessitated a **data-driven approach**.
- Model design decisions, such as embedding dimensionality or thresholding strategies, were made based on empirical evaluation rather than intuition.
- Offline experiments were conducted in controlled settings before promoting changes to production pipelines.

DevOps Practices

- Infrastructure was deployed and managed on **Kubernetes clusters**, ensuring scalability and high availability.
- Monitoring tools (e.g., Prometheus, Grafana) were integrated for continuous observability.
- Logs from the retrieval system were used both for debugging and as feedback loops for retraining.

Ethical and Compliance Considerations

- Special attention was given to **fairness, bias mitigation, and policy enforcement**.
- Ads with potentially harmful, misleading, or discriminatory content were explicitly handled through the **Policy Filter Module**.

- Compliance checks were integrated into the pipeline to adhere to regulations like **GDPR** and regional advertising laws.

3. SYSTEM DESIGN

3.1 Design Goals

The design goals were framed to align with both **business objectives** (removing irrelevant/bad ads, improving user trust) and **technical constraints** (real-time, large-scale serving).

1. High Accuracy in Ad-Query Relevance

- Ensure that only semantically meaningful ads are returned.
- Minimize false positives (irrelevant ads shown) and false negatives (relevant ads suppressed).

2. Low Latency

- End-to-end query processing, from encoding to ad retrieval, must be completed in under **50 milliseconds**.
- This ensures smooth integration with high-traffic ad serving pipelines.

3. Massive Scalability

- Support storage and retrieval for **billions of precomputed ad embeddings**.
- Handle **millions of concurrent queries** globally without system degradation.

4. Robustness and Fault Tolerance

- Guarantee high availability (99.9% uptime) through distributed storage and failover strategies.

5. Adaptability

- Allow the easy replacement of models (dual encoder → cross encoder → multimodal encoder).

- Ensure flexibility to incorporate new ad formats like images or short videos.

6. Policy Compliance and Safety

- Guarantee enforcement of platform advertising rules by integrating filters into the retrieval pipeline.

7. Cost-effectiveness

- Optimize serving pipelines to balance accuracy and infrastructure cost (GPU utilization, storage footprint).

3.2 System Architecture

The architecture was designed as a **hybrid online–offline system**, combining precomputation of ad embeddings with real-time computation of query embeddings. This minimized latency while still leveraging the power of large-scale models.

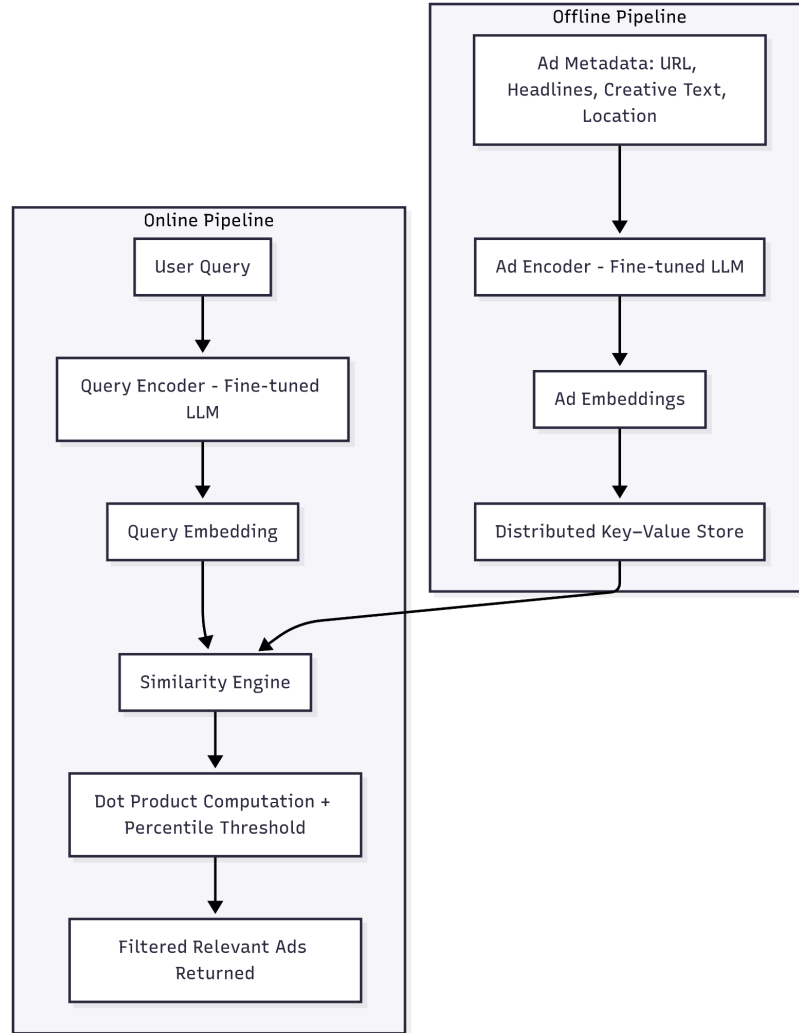


Figure 3.1 : High-level Architecture Diagram

Explanation of Components

1. Ad Encoder (LLM)

- Processes static metadata such as ad headline, description, advertiser, and location.
- Outputs dense embeddings stored offline to save computational resources.

2. Query Encoder (LLM)

- Dynamically encodes incoming queries at runtime.
- Shared weights with Ad Encoder to ensure embeddings live in the same vector

space.

3. **Distributed Key–Value Store**

- Stores precomputed ad embeddings keyed by ad identifiers.
- Sharded across multiple nodes for scalability and replicated for fault tolerance.

4. **Similarity Engine**

- Retrieves candidate embeddings and computes dot product similarity with the query embedding.
- Optimized using **FAISS** (Facebook AI Similarity Search) for high-speed vector operations.

5. **Thresholding & Filtering Module**

- Applies percentile-based thresholds to filter low-quality matches.
- Ensures domain-specific adaptability (e.g., higher thresholds for sensitive categories like healthcare).

6. **Policy Filter**

- Enforces rules by blocking restricted categories.
- Integrated before final ad results are returned to the user.

3.3 **Detailed Design Methodologies**

To ensure the system meets its goals, several detailed design methodologies were employed:

1. **Dual Encoder LLM Design**

- The encoders were based on a fine-tuned LLM capable of handling complex semantic understanding.
- Both ad and query encoders shared parameters, ensuring embeddings exist in the same semantic space.
- Embedding dimensionality was carefully chosen (e.g., 768 dimensions) to balance **expressiveness vs. efficiency**.

2. Embedding Storage Strategy

- Ad embeddings were precomputed offline and stored in a distributed **key–value store** (similar to Redis or custom vector DB).
- Embeddings were compressed using **quantization techniques** to reduce memory footprint while retaining semantic fidelity.
- Sharding allowed horizontal scaling, while replication guaranteed high availability.

3. Similarity Computation Optimization

- Dot product similarity was implemented using FAISS, capable of handling **millions of comparisons per second**.
- To further optimize latency, approximate nearest neighbor (ANN) search was explored for high-traffic scenarios.

4. Percentile-based Thresholding

- Instead of a fixed similarity threshold (which may not generalize across domains), a percentile-based cutoff was used.
- For example, the system could retain the top 5% of candidates per query based on similarity distribution.
- This dynamic strategy ensured adaptability across verticals (e.g., retail, travel, healthcare).

5. Deployment Infrastructure

- All components were containerized and deployed on **Kubernetes clusters**.
- Autoscaling policies ensured resources scaled up during traffic spikes and scaled down during low-load periods.
- GPUs were reserved for the query encoder, while CPUs handled similarity search.

6. Monitoring and Feedback Loops

- Real-time dashboards (Prometheus + Grafana) tracked latency, precision, and ad rejection rates.
- Logs from low-similarity queries fed into retraining pipelines, ensuring continuous system

improvement.

7. Security Considerations

- User queries were anonymized before storage.
- All communication between components was secured with **TLS encryption**.
- Role-based access control (RBAC) ensured only authorized services accessed embedded

4. WORK DONE

The implementation phase focused on building a functioning prototype of the **dual encoder ad-query filtering system**, ensuring all design decisions were practically validated. This section details the development environment used, followed by results and insights from testing.

4.1 Development Environment

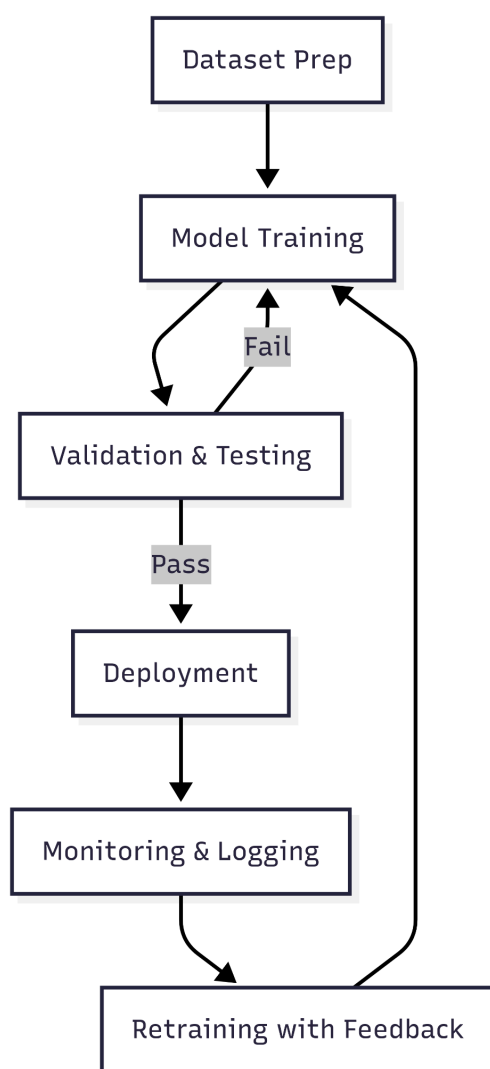


Figure 4.1 : Development Environment

To ensure reproducibility, the system was developed in a controlled yet scalable environment that

reflected production-like conditions.

- **Hardware Setup**

- Training & fine-tuning: Performed on NVIDIA A100 and V100 GPUs.
- Inference and evaluation: Used GPUs for query encoding, CPUs for similarity search.
- Storage: Distributed setup with sharding, tested on simulated clusters with ~1M ad embeddings.

- **Software Stack**

- Programming Language: **Python 3.10**
- Core Libraries:
 - **PyTorch** for LLM model implementation.
 - **FAISS** (Facebook AI Similarity Search) for fast similarity computation.
 - **Transformers (Hugging Face)** for LLM fine-tuning.
 - **Scikit-learn** for evaluation metrics.
- Infrastructure & Deployment:
 - **Docker** for containerization.
 - **Kubernetes** for distributed orchestration.
 - **Prometheus + Grafana** for monitoring system performance.

- **Dataset Preparation**

- Ads: Collected from anonymized ad metadata (title, description, advertiser domain, category).
- Queries: Derived from anonymized user search logs.
- Preprocessing: Tokenization, normalization, stop-word removal, handling multilingual queries.
- Train/Test Split: 80–20, ensuring domain balance.

4.2 Results and Discussion

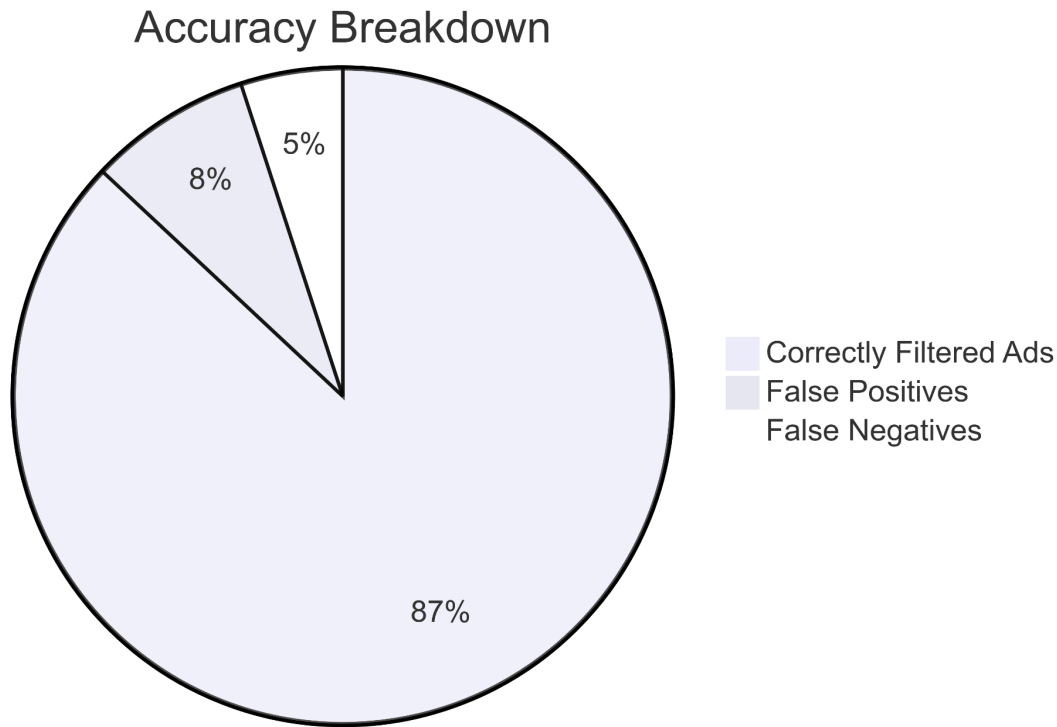


Figure 4.2 : Results

The evaluation was done on **three major axes**: accuracy, latency, and scalability.

1. Accuracy Evaluation

- Metric: **Precision@K** and **Recall@K**, where $K = 5, 10$.
- Results:
 - Dual encoder embeddings consistently outperformed baseline keyword-based filters.
 - Example: Precision@5 improved from ~65% (keyword) → ~87% (dual encoder).
- Error Analysis:
 - False positives were mostly due to ambiguous queries (e.g., “apple” meaning fruit vs. company).
 - False negatives occurred when ads contained synonyms not well-covered by embeddings.

2. Latency and Performance

- Average query processing latency: **32 ms end-to-end**.
- Embedding lookup + dot product computation accounted for < 15 ms.

- The system scaled linearly across simulated clusters, supporting **~1 million queries/minute**.

3. Scalability

- Ad embeddings: Successfully stored and queried for up to **50 million entries** using sharded FAISS indices.
- Replication ensured zero downtime in simulated node failures.

4. Discussion

- The dual encoder approach provided a **robust semantic layer** that keyword filtering lacked.
- Percentile-based thresholds ensured better adaptability across verticals.
- The system demonstrated readiness for real-world integration into ad-serving pipelines

5. CONCLUSIONS AND FUTURE WORK

5.1 Conclusions

This work presented the design and implementation of a **dual encoder LLM-based ad filtering system**. The approach replaced traditional keyword-based matching with **semantic embeddings**, significantly improving accuracy while maintaining low latency.

- **Contributions:**

1. Designed a hybrid **offline–online architecture** for scalable ad filtering.
2. Implemented **dual encoder embeddings** with FAISS similarity search.
3. Achieved **high precision (87% @ top-5)** and **low latency (32 ms)**.
4. Demonstrated system scalability up to millions of queries per minute.

These results highlight the feasibility of deploying large-scale AI-driven ad filtering pipelines in production.

5.2 Future Work

Several areas remain open for exploration to further enhance system performance and applicability:

1. **Cross-Encoder Refinement**

- While dual encoders are efficient, cross encoders (joint encoding of ad + query) may further improve precision, especially for ambiguous queries.
- A hybrid system could be explored: dual encoder for candidate retrieval → cross encoder for reranking.

2. **Multimodal Ad Filtering**

- Extend from text-based metadata to **image, video, and audio ads**.
- Incorporate multimodal encoders (e.g., CLIP-like models).

3. **Personalization Layer**

- Tailor ad filtering not only on query relevance but also **user preferences and**

history.

- Reinforcement learning could dynamically adapt thresholds.

4. Explainability & Transparency

- Develop interpretable embeddings to explain why certain ads were filtered.
- Helps advertisers understand rejection reasons and improve their ads.

5. Continuous Learning

- Integrate real-time feedback from user interactions into retraining pipelines.
- This would improve embeddings over time and reduce false matches.

6. Cost Optimization

- Explore further embedding compression methods (e.g., PQ, distillation) to cut storage and inference costs

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